

Future Scope & Architectural Roadmap

Project: LearnLens AI

1. Vision & Long-Term Evolution

LearnLens AI is designed to evolve from an intelligent lecture companion into a comprehensive **Personal Learning Operating System (PLOS)**. The foundational architecture—universal screen observation, multimodal fusion, session-scoped memory, and grounded pedagogical agents—provides a robust platform for future expansion.

2. Roadmap Phases

```

timeline
title LearnLens AI Future Evolution Roadmap
Phase 1 : Production MVP : Chrome DisplayMedia Capture : faster-whisper : dHash Deduplication : Session-scoped RAG : PPT & Dual PDF Generators : Human Control State Machine
Phase 2 : Advanced Agentic Pedagogy : Socratic Dialogue Loops : Code Execution Sandboxes : Automated Equation LaTeX Conversion : Multi-speaker Diarization
Phase 3 : Collaborative Knowledge Mesh : P2P Encrypted Study Group Sync : Anki / Notion Deep Sync : Local Vector Cross-Indexing : Offline Mobile Companion
Phase 4 : Ambient Educational OS : Multi-monitor Learning Tracking : AR/VR Spatial Slide Projection : Adaptive Spaced Repetition Scheduling

```

3. Key Future Capabilities

3.1 Live Code Sandbox Integration

Allow the AI Teacher to extract live code snippets from captured programming lectures (Python, Rust, TypeScript) and spin up a secure, ephemeral WebAssembly container (Pyodide / WebContainers) allowing the student to execute and modify code directly inside the tutoring interface.

3.2 Spaced Repetition & External Knowledge Sync

Export extracted concepts and flashcards directly into Anki (`.apkg`) and Notion databases via official local APIs, scheduling automated active-recall notifications based on Ebbinghaus forgetting curve intervals.

3.3 Peer-to-Peer Encrypted Study Groups

Enable students in the same university cohort to securely synchronize session knowledge bases using decentralized local mesh protocols (e.g. Libp2p / WebRTC) with zero central server storage, preserving student privacy while crowdsourcing lecture slide clarification.